

Analogue Electronics for Capacitive Sensors

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Abstract—This study presents the design and implementation of an analogue electronic readout circuit for a capacitive pressure sensor. The AC signal generated by the designed oscillator passes through the capacitive pressure sensor, which replaces the input capacitor, and reaches the amplifier input. Changes in the sensor capacitance modify the amplifier input, thereby changing the output voltage. A DC voltage proportional to the output signal is obtained using a half-wave rectifier.

Index Terms—Analogue readout circuit, capacitive pressure sensor, oscillator, transistor-level amplifier, AC-DC conversion.

I. INTRODUCTION

A. Purpose of the project

Capacitive sensors are widely used for detecting mechanical quantities. As Puers emphasizes, the fundamental limitation of capacitive sensors is that they do not provide a direct voltage or current output; instead, they create changes in capacitance. To read these changes, an analog readout circuit is required. In the circuit used, the AC signal generated by the Wien bridge oscillator is connected to the base of a transistor-level amplifier through a capacitive pressure sensor that replaces the input coupling capacitor. Pressure-induced capacitance changes alter the amplifier input and thus the output signal. This signal is then converted to DC voltage via an AC-DC conversion stage. The system has been designed, simulated, and experimentally characterized as a capacitive pressure sensor readout device [1] [1]

B. Application areas

Capacitive pressure sensors differ from other sensing technologies due to their low power consumption, high sensitivity, and long-term stability. One of the most critical application areas of capacitive pressure sensors is biomedical systems. Examples include intracranial pressure monitoring, intraocular pressure sensors, and blood pressure measurement devices [1]. Capacitive pressure sensors are also employed in wearable devices for respiratory monitoring, where changes in capacitance are caused by moisture variations in human breath.

In addition to biomedical applications, capacitive pressure sensors are commonly used in industrial pressure monitoring and process control systems. These sensors are widely applied in environmental monitoring applications such as atmospheric pressure and airflow measurements. Furthermore, capacitive pressure sensors are extensively utilized in microelectromechanical systems (MEMS), where compact size and high

accuracy are required. They are also used in automotive and embedded systems for applications such as tire pressure monitoring, airbag systems, active suspension control, and engine manifold pressure measurement [2].

II. DESIGN CONCEPT INCLUDING BLOCK DIAGRAMS

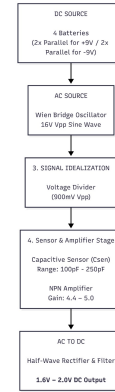


Fig. 1. Block diagram of system.

III. WORKING PRINCIPLE (INDIVIDUAL PARTS) INCLUDING CIRCUIT DIAGRAMS

A. Dc Source

In the oscillator, the op amp normally operates with a 9 V power supply. When setting up the circuit on a breadboard, a 9 V battery was used as a portable power source. For sufficient current capability, two 9 V batteries were connected in parallel. In a parallel battery configuration, the supply voltage remains unchanged,

$$V_{\text{total}} = V_1 = V_2, \quad (1)$$

while the available current increases as the sum of the individual battery currents,

$$I_{\text{total}} = I_1 + I_2. \quad (2)$$

The positive supply, negative supply, and ground connections of the circuit were provided through these two parallel-connected batteries.

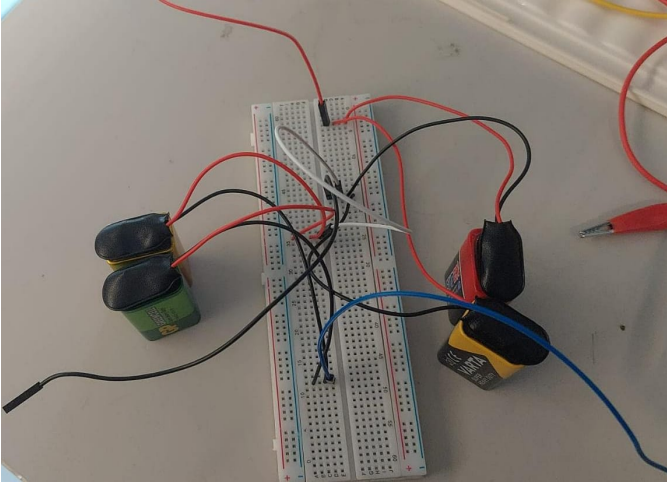


Fig. 2. Dc source on breadboard

B. AC Source

Sinusoidal oscillators consist of amplifiers containing RC or LC circuits. Op-amp sine-wave oscillators operate without an externally applied input signal. Positive and negative feedbacks are used to put the op-amp into an unstable state which causes the output to swing back and forth between positive and negative. Consequently, the Wien Bridge was selected primarily for its stability and ability to generate pure sinusoidal waves. It also differs from other oscillator types with its ease of frequency adjustment by the RC circuit. In the Wien bridge, positive feedback determines the frequency, while negative feedback control the gain.

In the positive feedback path, the oscillation frequency is given by

$$f = \frac{1}{2\pi\sqrt{R_1 C_1 R_2 C_2}}. \quad (3)$$

If $R_1 = R_2$ and $C_1 = C_2$, the expression reduces to

$$f = \frac{1}{2\pi RC}. \quad (4)$$

Using $R_1 = R_2 = 5.6 \text{ k}\Omega$ and $C_1 = C_2 = 2.2 \text{ nF}$,

$$f = \frac{1}{2\pi(5.6 \times 10^3)(2.2 \times 10^{-9})} = 13 \text{ kHz}. \quad (5)$$

As a result, the oscillation frequency was set to 13 kHz specifically to lower the high capacitive reactance of the pico-farad level sensor, ensuring sufficient signal to the amplifier. The input signal is obtained directly from the feedback loop. In this two stage RC oscillator, the RC circuit weakens the signal to one-third of its original value. It means

$$\beta = \frac{1}{3}. \quad (6)$$

Therefore, for sustained oscillations,

$$|A\beta| \geq 3 \quad (7)$$

so the amplifier gain must be set to at least 3.

The negative feedback loop of the amplifier is implemented using a voltage divider circuit. For the non-inverting configuration employed in this design, the closed-loop voltage gain is determined by the ratio of these feedback resistors according to the standard equation

$$A_v = 1 + \frac{R_{f1}}{R_{g1}}. \quad (8)$$

At the end of this calculation,

$$A_v = 1 + \frac{15.3 \text{ k}\Omega}{5.6 \text{ k}\Omega} = 3.73. \quad (9)$$

At the end of the Wien bridge oscillator resistive voltage divider is used to scale down the signal amplitude for suitable level for amplifier. The voltage divider consists of two resistors connected in series, where the output voltage is taken from the midpoint between the resistors.

The relationship between the input and output voltages of the divider is given by

$$V_{\text{out}} = V_{\text{in}} \cdot \frac{R_{\text{lower}}}{R_{\text{upper}} + R_{\text{lower}}}. \quad (10)$$

In the implemented circuit, the upper resistor is chosen as $R_{\text{upper}} = 100 \text{ k}\Omega$ and the lower resistor as $R_{\text{lower}} = 5.6 \text{ k}\Omega$. With an input signal amplitude of approximately 16 V, the output voltage becomes

$$V_{\text{out}} = 16 \cdot \frac{5.6}{100 + 5.6} \approx 0.85 \text{ V}. \quad (11)$$

So, the voltage divider reduces the signal amplitude to approximately 900 mV. In conclusion, an AC power source created to supply power to the sensor.

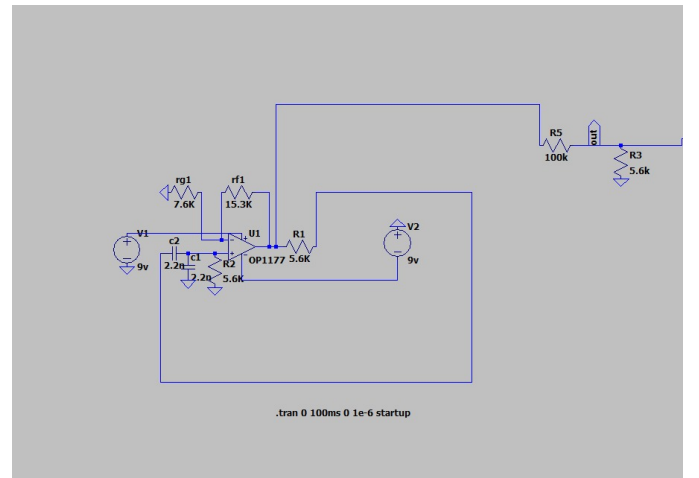


Fig. 3. Oscillator circuit schematic

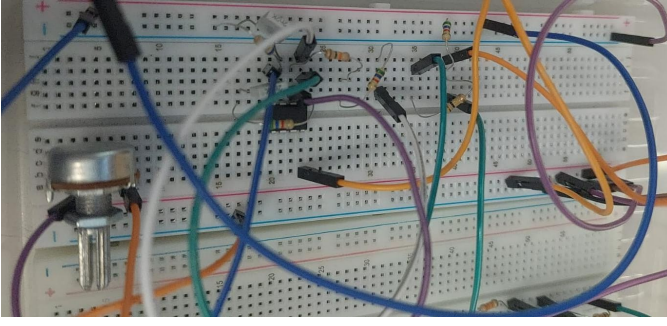


Fig. 4. Oscillator circuit on the breadboard

C. Amplifier

The amplifier operates in a common emitter configuration, where the signal applied to the base of the transistor is amplified to a higher voltage at the collector. Signal that oscillator output, together with the voltage-divider network, is goes to the amplifier input through R_1 before reaching the capacitive pressure sensor. In this circuit, the capacitive pressure sensor is placed at the amplifier's input and functions as a coupling capacitor. When pressure is applied, the sensor capacitance changes, which in turn changes how much of the AC input signal reaches the transistor base. An additional capacitor is connected in parallel with the capacitive pressure sensor to increase the effective input capacitance and thereby reduce the capacitive reactance at the amplifier input. The capacitive reactance is given by

$$X_C = \frac{1}{2\pi fC}. \quad (12)$$

Since the sensor capacitance is very small, it presents a high impedance to the AC signal. Adding a fixed capacitor in parallel increases the total capacitance and allows the signal to pass more easily to the amplifier. So the total capacitance increases according to

$$C_{\text{total}} = C_{\text{sensor}} + C_{\text{parallel}}. \quad (13)$$

The capacitive reactance decreases according to Eq. (10). This reduces the opposition to the AC signal, allowing a stronger signal to reach the amplifier input. At the same time, the sensor capacitance can still change with pressure, so the input signal remains pressure-dependent. A voltage-divider bias network formed by resistors R_2 and R_1 provides a constant DC voltage at the transistor base, establishing a stable quiescent operating point (Q-point). The Q-point represents the DC collector current and the collector-emitter voltage of the transistor in the absence of an input signal and ensures that amplification occurs in the active region. Small-signal AC operation takes place around this operating point. Variations in the input signal at the base terminal cause corresponding changes in the collector current, which are converted into amplified voltage variations across the collector resistor R_{C1} . For AC signals, the emitter bypass capacitor C_6 effectively shorts the resistor R_4 , leaving R_{E1} as the dominant emitter resistance. As a result, the voltage gain is primarily determined by the ratio

of R_{C1} to R_{E1} . Finally, the output coupling capacitor C_7 blocks the DC component and transfers only the amplified AC signal to the load resistor R_6 . This configuration enables stable amplifier operation across different sensor capacitance values, and an increase in sensor capacitance results in a corresponding increase in the output signal amplitude.

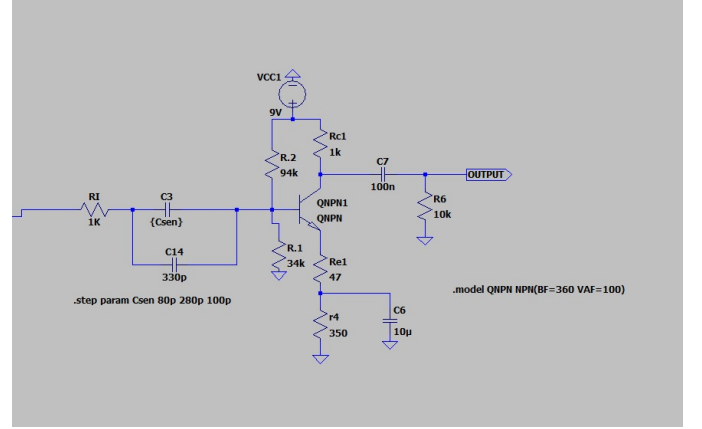


Fig. 5. Amplifier circuit schematic

General capacitive pressure works with 1 flexible diagram between 2 conductor plates. When the pressure increases diagram gets smaller than it causes the decreasing distance between 2 conductor plates. The formula of capacitance indirectly depends on the distance.

$$C = \epsilon_r \epsilon_0 \frac{A}{d} \quad (14)$$

In this project, this application is applied on the different ways. For the conductor plates, 2 copper plates are used and a piece of paper folded in half is put between them. To provide pressure, 2 different weight stones is used but surface area is chosen a plastic cup which has 9 cm length and 11cm with. The force is applied to pressure sensor is changed according to stones weight.

$$P = \frac{F}{A} \quad (15)$$

$$W = m \times g \quad (16)$$

The sensor's capacitance with no pressure is measured approximately 90pF. And when the first stone which is 2.4kg is put on the sensor. Capacitance change is observed over than 100pF. And this change is kept continuo as increasing of weight on the sensor.

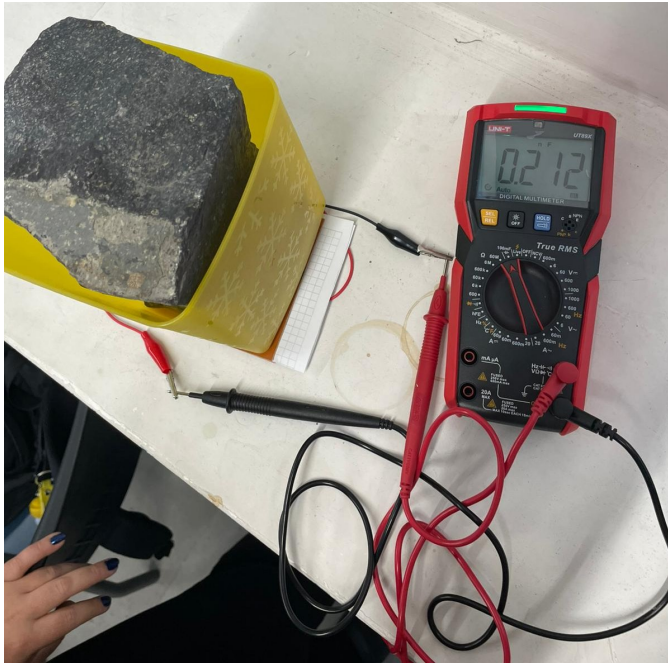


Fig. 6. Experimental setup of the pressure sensor and measurement circuit.

D. AC TO DC Converter

The diode allows only the positive half-wave of the AC signal to pass through, enabling the circuit to function as a half-wave rectifier. The RC filter following the diode filters the signal. The capacitor charges with the voltage coming through the diode and discharges through the resistor. Because the time constant is greater than that of the incoming signal, the capacitor begins to recharge before it can fully discharge. This results in a smooth DC voltage, except for minor ripples. This DC output voltage is related to the amplitude of the amplified AC signal and reflects the capacitance changes of the sensor due to pressure.

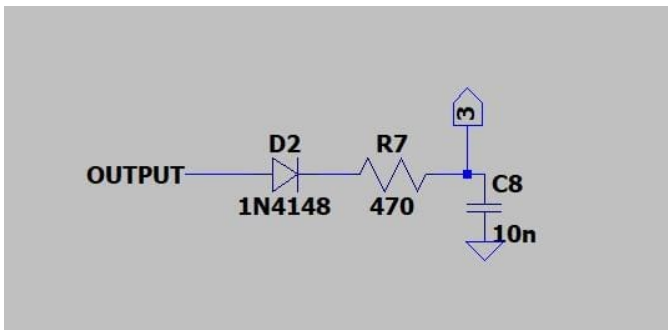


Fig. 7. Ac to dc circuit schematic

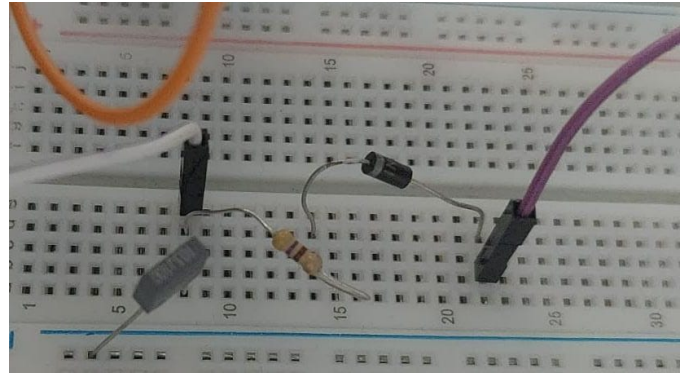


Fig. 8. Ac to dc on breadboard

IV. SIMULATION RESULTS (INDIVIDUAL PARTS AND THE WHOLE CIRCUIT)

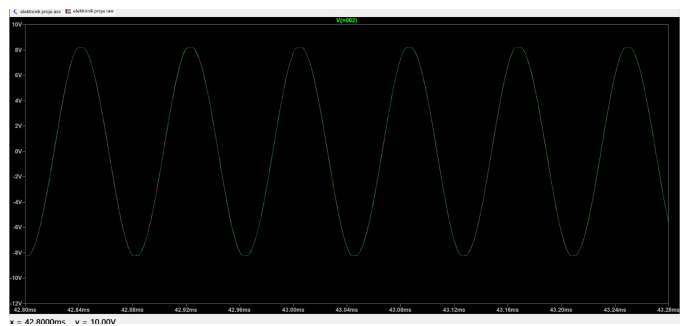


Fig. 9. Output of the Wien bridge oscillator

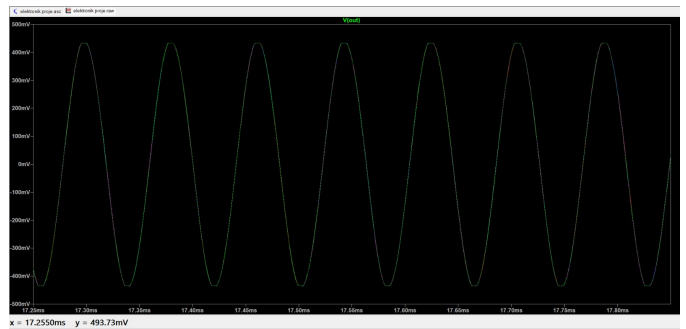


Fig. 10. Oscillator output signal after applying voltage divider

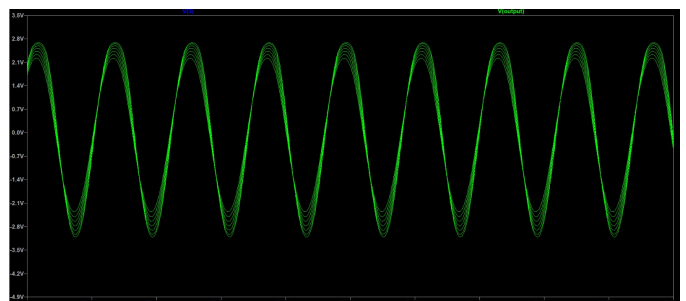


Fig. 11. Amplifier output

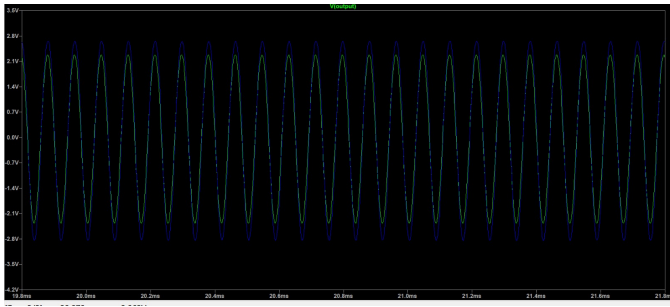


Fig. 12. Amplifier output for minimum and maximum sensor value

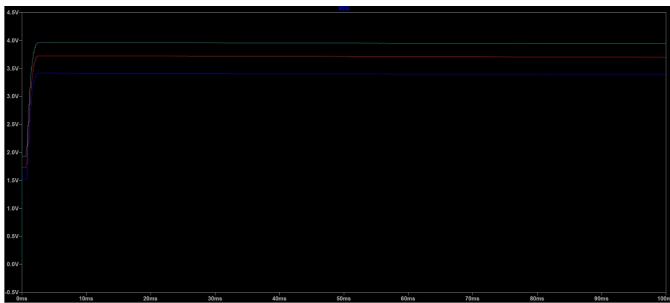


Fig. 13. Ac to dc output

V. EXPERIMENTAL RESULTS (INDIVIDUAL PARTS AND THE WHOLE CIRCUIT)

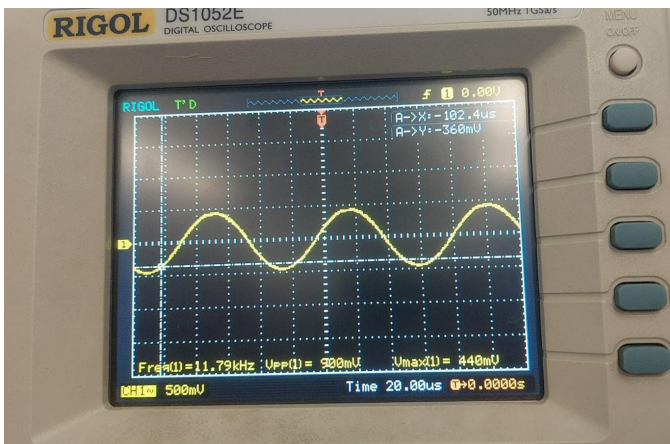


Fig. 14. Osillator output signal after Wien bridge and voltage divider

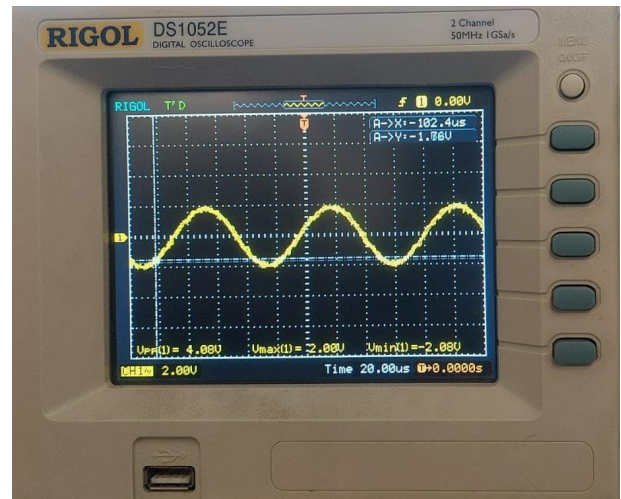


Fig. 15. Amplifier output signal when sensor is not full

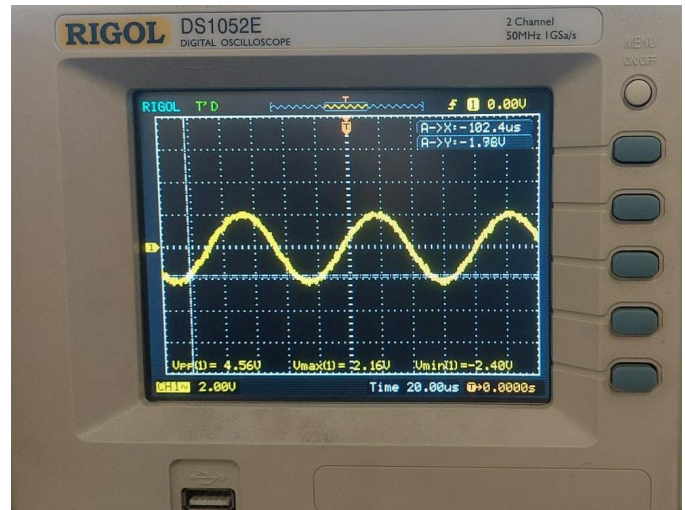


Fig. 16. Amplifier output signal when sensor is full

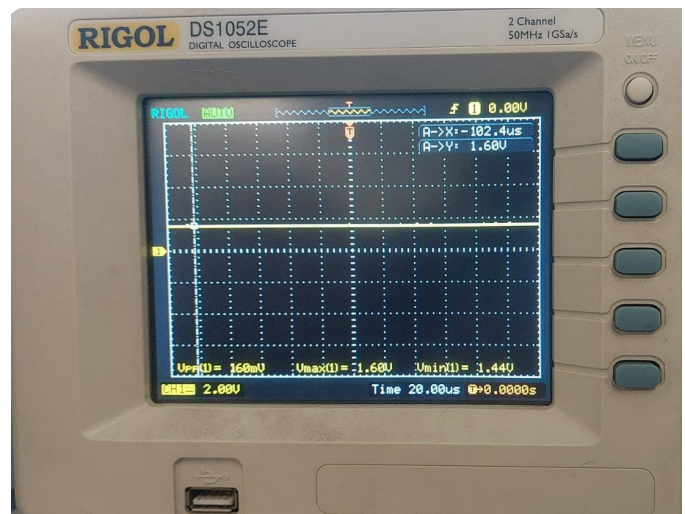


Fig. 17. Rectifier output when sensor is not full

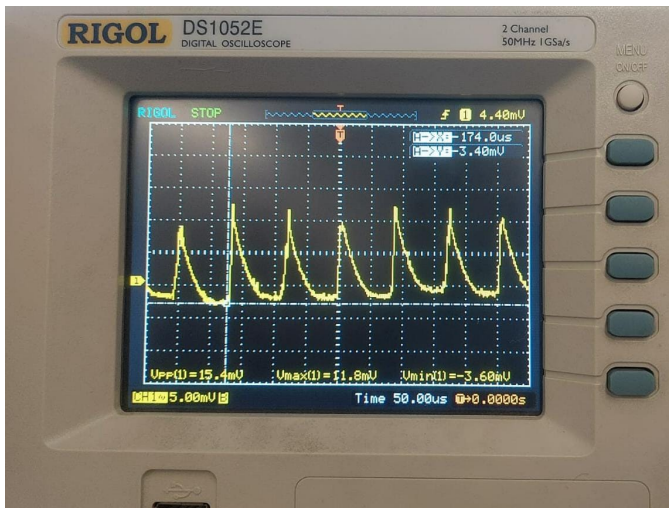


Fig. 18. Rectifier ripple output signal when sensor is not full

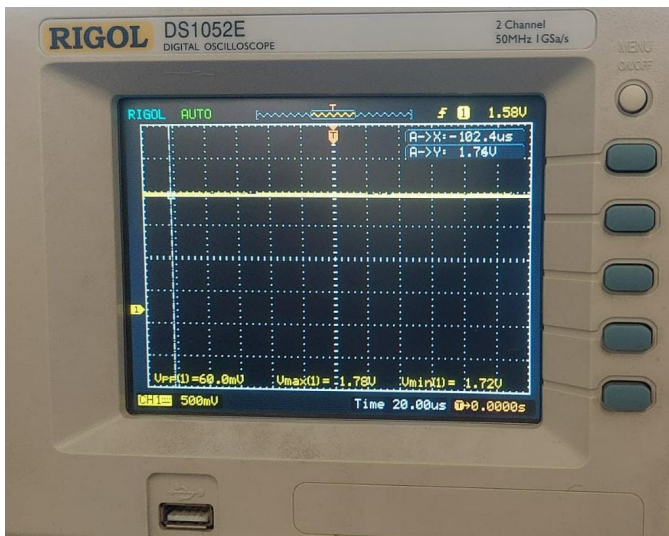


Fig. 19. Rectifier output when sensor is full

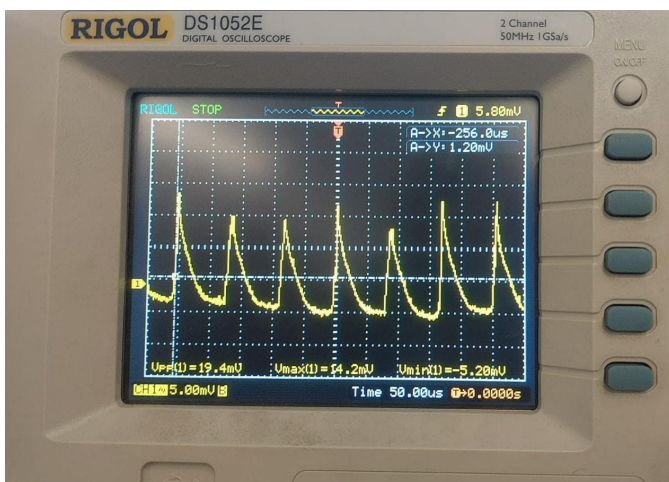


Fig. 20. Rectifier ripple output when sensor is full

VI. DISCUSSION

VII. CONCLUSION

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- [1] R. Puers, "Capacitive sensors: when and how to use them," *Sensors and Actuators A: Physical*, vol. 37–38, pp. 93–105, Jun. 1993, doi: 10.1016/0924-4247(93)80019-D.
- [2] J. Fraden, *Handbook of Modern Sensors: Physics, Designs, and Applications*, 5th ed. New York, NY, USA: Springer, 2016.